

# Urban Wage Premium\*

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## Abstract

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\*People to thank...

# 1. Introduction

Workers in urban areas earn a 16 percent wage premium over their non-urban counterparts. This gap is often referred to as the urban wage premium, and is a well documented fact in the urban economics literature (see for example Rosenthal and Strange (2004) for a review). The literature has closely explored the drivers of this gap, which include the role of sorting and selection of high productivity individuals into cities (Combes et al., 2008). For example Yankow (2006) documents that up to two-thirds of the urban wage premium is driven by differences in worker skills. The literature has also closely studied the role of differential returns to experience in urban areas as a source of the wage premium (Baum-Snow and Pavan, 2012). Outside of human capital explanations, researchers have explored the role of agglomeration economies and firm survival to explain the remaining gap (Glaeser and Maré (2001), Wheaton and Lewis (2002), Combes et al. (2012), Behrens et al. (2014)).<sup>1</sup> While the importance of sorting, selection, and firm survival are well documented in recent data, less is known about the urban wage premium throughout the 20th Century, and how the role of sorting and selection varied over time. This gap in the literature exists largely because there are few (if any) historical individual-level longitudinal datasets with income data, making it empirically challenging to fully account for individual sorting. Boustan et al. (2013) provide descriptive evidence of the rising urban wage premium throughout the 19th and 20th Century, which peaked around 1990, however, they do not isolate selection and sorting effects in their exercise. In this paper, we explore how the urban wage premium has evolved over the second half of the 20th Century and into the 21st Cen-

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<sup>1</sup>Recently, work has also shown that urban industry-level concentration has led to a markdown of wages as firms exercise monopsony power Azar et al. (2020), thus, in the absence of concentration, the urban wage premium may be even larger than previously thought.

tury, while addressing unobservable sorting following recent econometric advances introduced by Altonji and Mansfield (2018).

In our empirical exercise, we first document the raw urban wage premium over time. We then consider how much of the gap is explained by differences in observable individual characteristics. Finally, we seek to understand to what extent spatial sorting on unobservables, such as choice over unobserved amenities or unobserved productivity, drives the urban wage premium. Empirically, addressing sorting is challenging in the time period we explore due to a lack of individual level panel data. To net out the role of spatial sorting, we follow Altonji and Mansfield (2018) and control for MSA level averages of individual characteristics. Intuitively, Altonji and Mansfield (2018) highlight that group averages of observables tell us how individuals optimally select or sort into locations, if one is willing to assume that average observables are correlated with group averages of unobservables. Thus, controlling for observable group averages controls for many unobservables that drive selection and sorting. Adopting this approach allows us to document the role of sorting on the urban wage premium prior to the construction individual-level panel datasets such as the NLSY or PSID, which are used in recent works by Baum-Snow and Pavan (2012), Glaeser and Maré (2001), and Yankow (2006)).

Consistent with prior literature, we document a large urban wage premium in the raw data, ranging from 20-35 percent over the postwar era. The raw urban wage premium peaks in 1940, and declines rapidly between 1940 and 1960, and then declines more slowly throughout the period after 1960. We also see that the unadjusted premium rises again after 1980, which closely follows the skills biased technological change literature (see Acemoglu and Autor, 2011 for a survey). Controlling for observable individual characteristics reduces the urban wage premium. We find that individual characteristics explain XX - YY of the urban wage premium, a much smaller

share than Yankow (2006) describes, albeit, we have a much more limited set of observables due to the nature of our data. When we add controls for MSA-level averages of individual characteristics and MSA average characteristics, following Altonji and Mansfield (2018), we find that the urban wage premium is about 1/3 the size of the estimate that only controls for individual observables. Importantly, we see that after netting out the role of spatial sorting, the urban wage premium drops rapidly between 1940 and 1960, and then levels off to about 5 percent throughout the remainder of the 20th and 21st Century. Our estimate of 5 percent in the period 1960-2015 is very similar to the estimates reported in Baum-Snow and Pavan (2012), Glaeser and Maré (2001), and Yankow (2006) that use individual fixed effects models. The estimates prior to 1979 are new to the literature.

This raises an interesting question, what explains the decline in the urban wage premium in the postwar era, or alternatively, why did spatial sorting become increasingly important? In the second half of the empirical exercise, we explore whether urban-rural convergence explains the pattern that we documented above. We follow the approach of Barro and Sala-I-Martin (1990,1991) and estimate the rate at which MSAs and non-MSAs converge over the second half of the 20th Century. A large literature has explored the convergence of incomes in the United States over the 20th Century (East-erlin, 1960, Liebergott, 1966, Wright, 1987, McClean and Mitchener, 1999). However, to our knowledge, no work has explored the convergence of the urban wage premium over time. We estimate that rural-urban wage convergence occurs at a rate of XXX between 1940 and 2015. We also consider the rate of convergence between different periods (1940-1960, 1960-2015) and find that it is fastest between XXX and YYY. Importantly, we also explore what factors led to differential convergence rates over time. In particular, a large literature has explored the role of structural transformation in the United States (Caselli and Coleman, 2001), demographic changes stemming

from the Great Migration (Boustan 2016, Collins and Wannamaker, 2017, Derenoncourt, 2022), and growing agglomeration forces (Krugman and Venables, 1995). We follow Ganang and Shoag (2017) and highlight how the urban wage premium differentially converges in locations with different initial conditions along dimensions of initial industrial structure, demographics, and economic integration via the highway network.

We find that the rural-urban premium converged from below in locations with initially low levels of manufacturing and from above for places with initially high levels of manufacturing. This pattern is consistent with Caselli and Coleman's (2001) pattern of structural transformation. We also find that location with initially high shares of black population converged from below. This is consistent with the Great Migration and structural transformation as blacks left agriculture in the South to seek employment in northern industrial areas. Finally, we find that locations that had initial Market Access advantages converged from above, which is consistent with the South and West experiencing relatively larger transportation improvements with the construction of the Interstate Highway System in the 1960s.

A remaining question is why the urban wage premium has not fully converged. We follow Ganaong and Shoag (2017) and examine the role that land use regulation has on the convergence of the urban wage premium. We find that places where land use regulation is relatively low have converged, while locations with high levels of land use regulation have not. This finding highlights how inelastic housing supply can generate inefficiencies in the spatial distribution of employment.

## **2. Historical Background**

## 2.1 The Rise of Cities

The formation of cities can reduce the cost of transporting ideas, goods, and people through colocation (Marshall, 1920). At the same time, increasing density can create congestion externalities, such as traffic congestion, or, as documented in early American history, a worse disease environment. In the United States, cities initially formed in places where it was convenient to trade and process agricultural goods due to natural advantages (Boustan et al., 2013). As canals expanded in the early 19th Century, new cities sprang up to support the shipment and transshipment of goods. Later, the railroads bypassed many previous cities and led to new urban centers.<sup>2</sup> With the arrival of the railroads and improved mechanical technology, the size and scope of firms changed, leading to new industrial cities (Chandler, 1959, 1993). These highly productive cities, mostly located in the North and Midwest contributed to the high wages experienced outside of the South (Wright, 1986). Throughout the 19th Century there were substantial wage differentials between regions in the United States (Wright, 1986, Rosenbloom, 1990, Kim and Margo, 2004). During the 20th Century, wages have largely converged across regions, but there remain large gaps between urban and rural areas (McClean and Mitchener, 1999).

What explains the wage gap between regions and between rural and urban areas? On the labor supply side, urban economists have shown that city sizes, wages, and rents will be determined by a spatial equilibrium whereby workers move until utility is equalized between locations (Rosen, 1979, Roback, 1982). If rural and urban locations are fully integrated, real wages should equalize across locations. Alternatively, any nominal wage difference will be offset by higher prices (rents) due to higher productivity or better amenities. Therefore, any differences in the real wage may reflect migration frictions

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<sup>2</sup>As Bleakley and Lin (2012) demonstrate, agglomeration forces allow cities to exist after their technological obsolescence.

that prevent workers from arbitraging the higher wage in cities. These frictions can arise from a variety of sources, moving or job switching costs (Ransom, 2022), incomplete credit or insurance markets (Morten, 2019), strong local preferences or local familial ties (Diamond, 2016, Zabek, 2020), or other factors.

At the same time, firm behavior may also explain the urban wage premium. As Glaeser and Maré (2001) note, it is puzzling that firms would remain in locations where wages are high. In a spatial equilibrium framework, if firms remain in the urban location, then it must be the case that there are productivity differences between urban and rural locations. Of course, not all firm capital can be redeployed and is costly to relocate (Langenmayr and Simmler, 2021). Fully disentangling imperfect mobility from differential productivity is empirically challenging. Adopting recent econometric techniques enables us to address endogenous labor sorting, which can address differences in the composition of labor if workers sort on amenities or specific skills. However, net of labor sorting, our approach does not permit us to differentiate between differences in urban productivity or migration frictions. Our measure of the urban wage premium will reflect the combined effect of unmeasured productivity and migration frictions.

Understanding how the wage premium has evolved over time provides insight to the changing patterns of sorting and to what extent rural and urban economies are integrated with one another. In the period that we study, 1940-2015, there have been several major changes in the economic structure of the United States that affected the relative attractiveness of locations and migration costs. First, both biological and mechanical innovations in agriculture altered patterns of crop specialization and the optimal mix of labor and capital, freeing both land and labor for alternate uses (Caselli and Coleman II (2001), Michaels et al. (2012), Olmstead and Rhode (2001), Griliches (1963), and Jung (2020)). There were also drastic changes in transportation,

such as the improved highway network, that have facilitated the growth of cities and reallocated population across space. For example, Rephann and Isserman (1994) and Duranton and Turner (2012) highlight the role of the interstate highways in expanding growth rates and city-level employment in the United States. At the same time, as cities have grown, so too has the regulation of land use, especially following *Euclid vs. Ambler* (1926), which may increase frictions for individuals who desire to move to urban areas to take advantage of higher wages.

## 2.2 Highway Background

## 3. Data

To estimate the urban wage premium throughout the post war era, we draw on several datasets. First, we draw on publicly available microdata of decennial Censuses and the American Community Survey (ACS). To measure the average earnings per worker (wages), we use the IPUMS Full Count Census for 1940, the 1 percent sample Census for 1950 and 1970, the 5 percent sample Census for 1960, 1980, 1990, and 2000, and the 2005, 2010, and 2015 American Consumer Surveys. Our sample consists of adult men living in the continental US between the ages of 25 and 65 who are employed for at least 26 weeks in the sample year. We apply IPUMS person weights in our summary statistics and all our regressions.

We compute average weekly earnings (wages) by finding total income divided by the number of weeks worked in the given survey year. For individuals earning the top income code, we set their total income to the lower top income cutoff multiplied by 1.5.<sup>3</sup> If the urban wage premium is primarily

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<sup>3</sup>These cutoffs are reported in IPUMS INCWAGE documentation. In 2015 \$, the cutoffs are \$84,500 for 1940, \$99,200 for 1950, \$199,500 for 1960, \$309,000 for 1970, and \$225,000 for 1980. Afterwards, IPUMS reports income as the median wage above that year's cutoff.

driven by people in the top income group, our estimates of the urban wage premium will be biased downwards by this choice. All incomes are then multiplied by the Consumer Price Index to be in 2015 dollars. In order to have a consistent measure of occupation, we use the ‘occ1950’ variable provided by IPUMS which harmonizes occupation codes across all our survey years. In Table 1 we report the number of individual observations in each cross section.

Table 1 also reports the average weekly urban and nonurban wage in each time period. Two striking trends appear from the summary statistics. First, as we noted previously, the share of the population living in an urban areas has steadily increased between 1940 and 2015. This trend matches those highlighted in Boustan et al. (2013). Secondly, it is clear that both urban and nonurban mean weekly wages are growing over time in 2015 \$ from \$33.24 and \$25.23 in 1940 to \$1353.95 and \$1067.48 in 2015. The urban wage premium, the ratio of the urban to nonurban wage, follows a W shape. The urban premium starts at 0.32, and then falls sharply between 1940 and 1950, where it declines to a low of 19 percent in 1980. The decline from 1940 to 1950 is well documented in the literature, and is often referred to as the Great Compression (Goldin and Margo, 1992). Since 1980, the urban wage premium has increased; in 2015 the premium was 26 percent.

## **4. The Urban Wage Premium: *1940-2010***

### **4.1 Regression Framework**

Our goal in this paper is two-fold. First, we aim to describe the observed urban wage premium from a historical perspective. For this purpose, we estimate the urban wage premium from 1940 to 2015 by comparing average

Table 1: Summary Statistics

| Year | N          | % Urban | log(Market Access) |      | Urban Weekly Wage (2015 \$) |       | Nonurban Weekly Wage (2015 \$) |       |
|------|------------|---------|--------------------|------|-----------------------------|-------|--------------------------------|-------|
|      |            |         | Mean               | SD   | Mean                        | SD    | Mean                           | SD    |
| 1940 | 16,939,585 | 0.66    | 3.39               | 0.87 | 33                          | 20    | 25                             | 17    |
| 1950 | 68,457     | 0.69    | 3.38               | 0.67 | 72                          | 41    | 59                             | 33    |
| 1960 | 1,364,970  | 0.66    | 3.33               | 0.66 | 124                         | 77    | 100                            | 60    |
| 1970 | 280,338    | 0.75    | 3.42               | 0.70 | 205                         | 138   | 166                            | 105   |
| 1980 | 1,910,440  | 0.73    | 3.22               | 0.57 | 409                         | 284   | 341                            | 221   |
| 1990 | 2,262,149  | 0.74    | 3.12               | 0.46 | 701                         | 562   | 558                            | 405   |
| 2000 | 2,583,088  | 0.79    | 3.08               | 0.42 | 1,008                       | 1,015 | 767                            | 677   |
| 2005 | 530,737    | 0.79    | 3.08               | 0.42 | 1,143                       | 1,146 | 863                            | 712   |
| 2010 | 530,783    | 0.80    | 3.01               | 0.51 | 1,228                       | 1,210 | 961                            | 819   |
| 2015 | 561,934    | 0.81    | 3.00               | 0.52 | 1,353                       | 1,446 | 1,068                          | 1,011 |

*Notes:* Summary statistics weights individuals by their person weight. Weekly wage is reported in 2015 \$. Market access is calculated from solving  $MA_c = \rho \sum_d \tau_{cd}^{-\theta} MA_d^{-1} Y_d$ , where transportation costs,  $\tau_{cd}$ , are calculated as described by Equation ??.

weekly wages for people who live in urban and non-urban areas.<sup>4</sup> For each census survey year, we run the following basic regression:

$$\log(\text{Wage}_i) = \beta_0 + \beta_1 \text{Urban}_i + \varepsilon_i, \quad (1)$$

where  $\text{Urban}_i$  is an indicator if worker  $i$  resides in an urban area (as classified by the Census Bureau). Therefore, our estimate of  $\beta_1$  captures the percent difference in weekly wages earned by urban-residing workers relative to non-urban-residing workers. These estimates will serve as a baseline de-

<sup>4</sup>Weekly wages are adjusted for inflation using the BLS' consumer price index in each survey year.

scription of the differences in wages that we *observe* in our survey data and are descriptive and should not interpreted as the “causal effect” of living in an urban area.

One potential difficulty in comparing estimates across years is that the Census Bureau clasifies new (typically smaller) urban areas throughout the decades. There is evidence that the urban wage premium grows with city size and therefore we might expect to see a declining premium over time due to an expansion of urban area status.<sup>5</sup> In a robustness exercise in the Online Appendix, Figure B.1 reestimates our specification using only urban areas as classified in 1940. The results are very similar in this specification.

The second goal of this paper is to understand the different sources underlying the wage premium. The unadjusted urban premium reflects a combination of a true causal wage premium; differences in skills due to endogenous selection of labor into urban areas; differential returns to labor market experience; and other potential city labor amenities such as agglomeration economies. To better understand how these different channels impact the urban wage premium, we extend our regression specification to control for confounding variables. Specifically, we estimate a set of regressions for each census survey year as follows:

$$\log(\text{Wage}_i) = \beta_0 + \beta_1 \text{Urban}_i + X_i \delta + \varepsilon_i, \quad (2)$$

where  $X_i$  is a vector of control variables for worker  $i$ . To facilitate consistent comparisons across years, we use the same set of individual-level covariates in each survey-year. This limits the number of variables we include to a set of 5-year age bins, indicator variables for education level, and an indicator for being in a racial minority. Therefore our set of control variables are not able to control for all potential sources of endogenous sorting. For exam-

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<sup>5</sup>For example, see Glaeser and Maré (2001).

ple, *unobserved* individual attributes like motivation and aptitude can lead to sorting but not be controlled for by the econometrician (Yankow (2006) and Combes et al. (2008)). Thus, we do not provide a causal interpretation to the estimate of  $\beta_1$  in equation (2) either. The aim of this exercise is to determine the amount of the overall observed wage premium that these observable covariates are able to explain and how that may change over time.

In section 5, we aim to provide a causal estimate of the urban wage premium over time by controlling for *all* sources of endogenous sorting into cities. Since we do not observe the same worker in multiple survey years, we cannot follow previous papers in including individual-specific fixed effects. Instead, we follow the methodology recently proposed by Altonji and Mansfield (2018) which provides conditions under which controlling for group-averages of observed variables controls for *all* the unobserved covariates that we don't have individual-level data for.

## 4.2 Results

The results of estimating equation (1) is displayed in Figure 1. The urban wage premium starts around 35% in 1940, decreases to 25% between 1950 and 1970, and starting in 1980 the premium converges to where it stands today at around 20%. Our results match the estimates found in Figure 3 of Boustan et al. (2013) for estimates before 1980 while our estimates in later years are smaller than their estimates.<sup>6</sup> Our estimates are close to others

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<sup>6</sup>The reason for the difference in estimates comes from our use of a regression framework. The method in Boustan et al. (2013) estimates the following

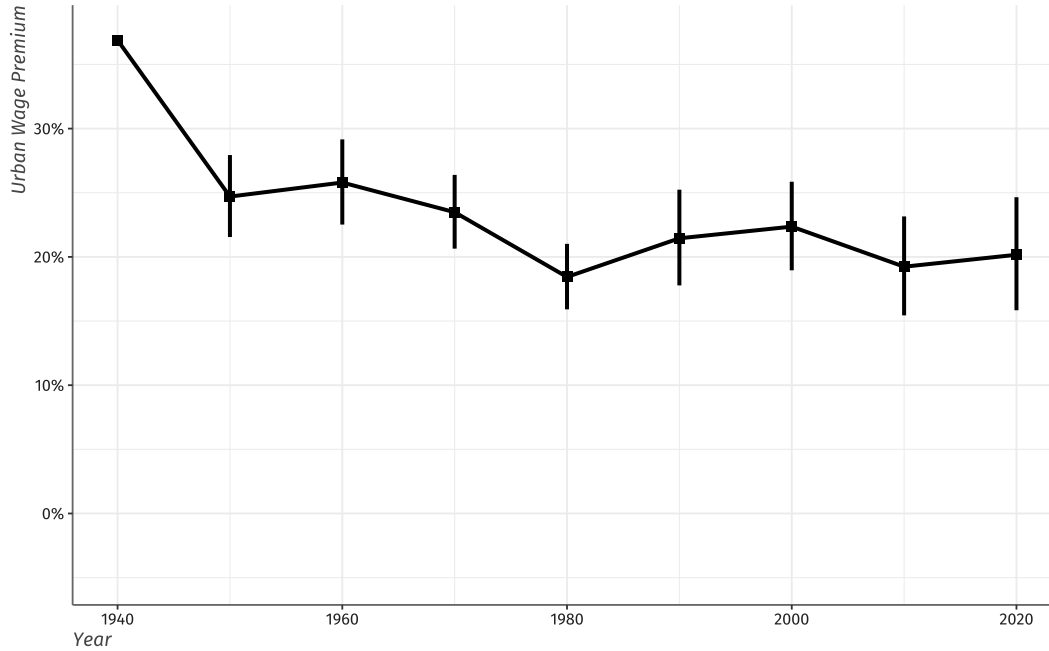
$$\log(\hat{\mathbb{E}}[\text{Wage}|\text{Urban} = 1]) - \log(\hat{\mathbb{E}}[\text{Wage}|\text{Urban} = 0])$$

which differs from the regression estimate

$$\hat{\beta}_1 = \hat{\mathbb{E}}[\log(\text{Wage})|\text{Urban} = 1] - \hat{\mathbb{E}}[\log(\text{Wage})|\text{Urban} = 0].$$

. The latter is a more precise estimate for average percent difference in wages.

Figure 1: Unadjusted Urban Wage Premium 1940-2015



Notes: Each point is the log of the ratio of average wage among urban male wage-earners and non-urban male wage-earners in the given survey year.

in the literature. Our unadjusted difference for the year 1990 and 2000 is smaller in magnitude than the estimate of between 21 and 32% from Glaeser and Maré (2001) and similar in magnitude to 1980 and 2000 as documented in Baum-Snow and Pavan (2012) and Boustan et al. (2013). Overall, the unadjusted estimates at the urban wage premium provide evidence for the overall decline in the ‘importance’ of cities for wages over time, but do not provide evidence of a ‘modern resurgence’ of the wage premium that is suggested by previous descriptive and historical work.

We report the urban wage premium estimates after controlling for individual characteristics in Figure 2 by the line with circle markers along with the original estimates marked with squares. Our estimates suggest that in 1940, the individual-level controls explain about 1/8 of the unadjusted ur-

ban wage premium, but the shape of the premium over time remains consistent. However, the explanatory power of individual controls steadily grows over time. In 2015, the individual-level controls explain about 1/3 of the urban premium highlighting the importance of sorting of individuals in explaining wage premiums in the modern era. To summarize, across all years in the sample, we find that individual-level observables explain between 8% and 33% of the urban wage premium, with the individual characteristics explaining a larger portion of the premium since 1980. However, due to our limited set of controls, these estimates should still be interpreted as descriptive.

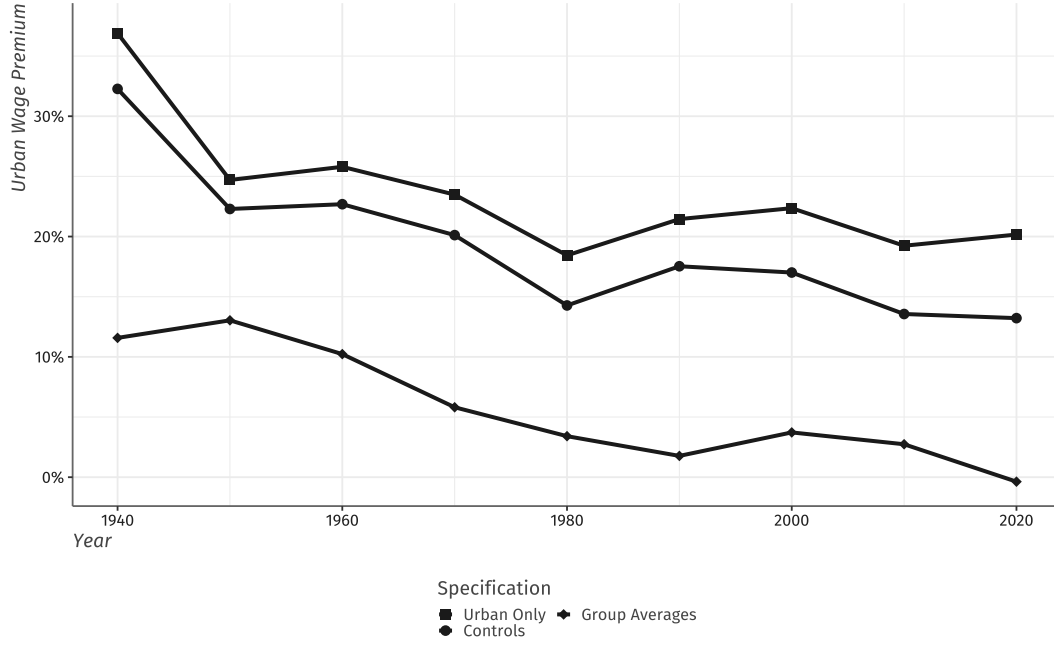
## 5. The Causal Effect of Urban Areas

### 5.1 Methodology

Given the cross sectional data that we employ, we cannot fully capture sorting by including individual fixed effects, a common strategy in the urban wage premium literature. Second, due to the many different datasets we use, we can only include a small set of harmonized covariates, thus we cannot reduce the scope of selection by controlling for additional covariates directly. Indeed, there are many sources of unobserved covariates that may lead to sorting between urban and non-urban areas that result in differences in average pay. To overcome this empirical challenge, we use the method proposed by Altonji and Mansfield (2018) to control for these unobserved variables driving selection.

Altonji and Mansfield (2018) show that under a set of conditions, there exists a *unique linear* mapping between group-level averages of *observed*

Figure 2: Urban Wage Premium 1940-2015



*Notes:* This figure plots the coefficient estimates for survey-year level estimates of the urban wage premium. In the grey line with square markers, the estimate is found by taking the ratio of the average log weekly wage for urban and nonurban workers. The remaining lines run versions of the regression given by Equation 2. In the line with diamond markers, there are no controls and no measure of market access; the line with triangle markers includes controls for age in 5-year bins, education levels, and an indicator for being White; the line with circle markers includes the same controls and the log of market access.

covariates and group-level averages of *unobserved* covariates.<sup>7</sup> Therefore including group-level averages linearly in Equation 2 will absorb *all of* the observed urban wage premium that is due to sorting on unobservable covariates.<sup>8</sup> The proposed method relies on two key assumptions regarding how individuals sort across locations. First, Altonji and Mansfield (2018) assume

<sup>7</sup>This is true even when there is no correlation between unobservable and observable variables at an individual-level.

<sup>8</sup>The group-level averages also will absorb the effects of the amenities that drive sorting, so changes in the wage premium coefficient reflect removal of both sources.

that individuals choose locations following a logistic choice model with a latent function that models the appeal of a location as a function of both the observed and unobserved covariates, a standard assumption in the location choice methodology. The second assumption, called the ‘rank condition’ requires that set of factors that drive sorting on *observable* covariates contains *all* factors that drive sorting based on *unobservable* covariates. Additionally, the rank condition requires that the number of group-average variables must be larger than the total number of factors driving sorting into urban areas.

The intuition for this approach is that the regression will first ‘invert’ the observed group-level averages to amenities, but this inversion is only possible if the mapping is one-to-one. If there are more amenities than group-average variables, then there is not be a unique inverse and the mapping between observable averages and unobservable averages breaks down. This second condition warrants more discussion as initially it may seem that there are many amenities that drive sorting in or out of urban areas. Diamond (2016) provides evidence using a principal component analysis on a very detailed set of amenities and documents that in the 21st century the effective rank of amenities is around 5 (with three components explaining most of the variation). Therefore, a key identifying assumptions is that our group-average variables have a rank larger than 5. Under these assumptions, we can then interpret our estimate of the urban wage premium as causal.

As an illustrative example, suppose that a part of the urban wage premium is driven by parent’s sorting into areas with high quality schools for their children. We may not be able to directly measure school quality (or some other amenity), but can observe the average education of parents in the location. So as long as the other parents’ location decisions are also correlated with unobserved school quality, controlling for the average parent education level will directly control for the sorting due to unobserved selec-

tion on school quality.

In the empirical specification, we include the following group-level averages, the share of individuals of different racial/ethnic groups, the share of individuals at different levels of educational attainments, the share of individuals who were veterans, the share of individuals of different marital status, and a measure of market access which summarizes a location ability to trade with other areas.

## 5.2 Results

The results following Altonji and Mansfield (2018)'s method are presented in Figure 2 by the line with diamond markers. First, we turn to estimates 1990 and later, as these are directly comparable to those reported in the modern urban wage premium literature. Under the assumptions laid out in Altonji and Mansfield (2018)'s, our estimates reflect the urban wage premium after controlling for sorting. We find an urban wage premium between 1990 and 2010 between 2% and 5%. These results fall very close to estimates in the literature that use panel data with individual fixed-effects to control for individual sorting. Glaeser and Maré (2001) estimate a premium of around 5% in the 90s; Yankow (2006) estimate a premium between 3-5% and D'Costa and Overman (2014) estimate a premium between 2-5% in the 2000s. Given that our estimates from 1990 to 2015 closely align with those found in the literature, we have confidence adopting this approach earlier in time.

Turning to the historical estimates, we estimate an urban wage premium of 25% in 1940. The wage premium falls rapidly between 1940 and 1960. In 1950, we estimate an urban wage premium of approximately 15%. In 1960, we estimate a wage premium of just 6%. This sharp decline in the urban wage premium is consistent with the Great Compression literature (Goldin and Margo, 1992) that highlights a narrowing of the wage gap im-

mediately following WWII. While the Great Compression literature points to a widening of the wage gap starting in the 1960s, we find that the urban wage premium is essentially flat and is roughly unchanged between 1960 and 2015.

We find this pattern of results fascinating. In part, the pattern of estimates shows that the raw premium present in 1940 was largely due to real differences in wages between urban and non-urban locations, whereas today, the raw premium is largely driven by differential sorting by high productive types. In 1940, sorting only accounted for (1-25/37)% of the premium, whereas in 2015, sorting accounted for roughly 75% of the raw wage gap.<sup>9</sup>

This raises an interesting question, what factors contributed to the closing of the urban wage premium in the second half of the 20th Century.

## 6. Convergence of the Urban Wage Premium

To better understand why the urban wage premium declined over time, we adopt the approach of Barro and Sala-I-Martin (1990,1991) and estimate the rate at which MSAs and non-MSAs converged over the second half of the 20th Century. From the estimates in the prior section, we can construct the relative premium for each MSA and non-MSA, which is equal to the average location wage minus the predicted average wage. This is a measure of how large the premium is for a given MSA is in a relative sense in each cross section. Following Barro and Sala-I-Martin (1990,1991), we can then estimate a regression as follows,

$$RP_{i,t} = \beta_0 + \beta_1 RP_{i,t-1} + \varepsilon_i, \quad (3)$$

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<sup>9</sup>In appendix XX, we show that the effects discussed here are not driven by changes in MSA classification.

where  $RP_{i,t}$  is the relative wage premium in the current period, in location  $i$ . If the relative premium has converged over time, then the coefficient,  $\beta_1$ , is less than zero. In this exercise, we consider the rate of convergence between different periods (i.e., 1940-2015, 1940-1960, 1960-2015).

## 6.1 Convergence Results

### 6.2 Drivers of Urban Wage Premium Convergence

An open question in the literature is what drove the convergence in wages between MSA and non-MSA's throughout the second half of the 20th Century. Here we explore several competing hypothesis by adopting the approach of Ganong and Shoag (2017) by modifying the basic convergence regression to allow a given factor  $C_{i,t0}$  to explain differential convergence rates. In general, we will estimate specifications of the following form:

$$RP_{i,t} = \beta_0 + \beta_1 RP_{i,t-1} + \beta_2 C_{i,t0} + \beta_3 RP_{i,t-1} C_{i,t0} + \varepsilon_i. \quad (4)$$

In this regression,  $\beta_3$  captures the differential rate of convergence due to the given factor.

We explore several different sources of change within the US during the second half of the 20th Century. First, there is a large literature that has explored the role of structural transformation in the United States on regional convergence (Caselli and Coleman, 2001). To address the role of structural transformation, we allow the convergence rate to differ by the initial share of manufacturing employment in the MSA. Caselli and Coleman (2001) describe the movement out of agriculture and into manufacturing, especially in the South. Therefore, locations with higher rates of manufacturing should experience urban wage premium convergence from above, as wages fall in response to in migration. Secondly, there were large demographic changes

in the United States in the early portion of our sample due in part to the Great Migration (Boustan 2016, Collins and Wannamaker, 2017, Derenoncourt, 2022). To address the role of the Great Migration, we allow convergence rates to differ by the initial share of the black population. Both the structural transformation of the US economy and the Great Migration are likely contributors to the Great Compression in wages observed 1940-1950, as described by Goldin and Margo (1992), Margo (1995), Jaworski and Niemesh (2018), and Collins and Niemesh (2019).

In post WWII era, the US invested in large scale highway infrastructure that may have altered agglomeration forces. The interstates in particular led to large gains in connectivity for locations outside of the East Coast and Rust Belt. We allow for differential convergence by including historic measures of market access. Finally, Ganang and Shoag (2017) highlight how convergence has stalled due to differences in land use regulation. We adopt their measure to test whether changes in land use explain the rate of convergence in the urban wage premium.

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## **A. Full Urban Wage Premium Regressions**

Full results of our regression estimates in Figure 2 can be found in Table A.1.

Table A.1: All Urban Wage Premium Estimates by Survey Year

|                                                                           | 1940               | 1950               | 1960               | 1970               | 1980               | 1990               | 2000               | 2005               | 2010               | 2015               |
|---------------------------------------------------------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| <b>Panel A: Urban only</b>                                                |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |
| Urban                                                                     | 0.3153<br>(0.0177) | 0.2208<br>(0.0131) | 0.2295<br>(0.0135) | 0.2109<br>(0.0119) | 0.1692<br>(0.0110) | 0.1943<br>(0.0157) | 0.2018<br>(0.0144) | 0.2014<br>(0.0160) | 0.1759<br>(0.0165) | 0.1661<br>(0.0175) |
| N                                                                         | 16,939,585         | 68,457             | 1,364,970          | 280,338            | 1,910,440          | 2,262,149          | 2,583,088          | 530,737            | 530,783            | 561,934            |
| <b>Panel B: Urban with Controls</b>                                       |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |
| Urban                                                                     | 0.2808<br>(0.0170) | 0.2012<br>(0.0125) | 0.2045<br>(0.0123) | 0.1833<br>(0.0119) | 0.1334<br>(0.0116) | 0.1615<br>(0.0162) | 0.1571<br>(0.0125) | 0.1517<br>(0.0131) | 0.1272<br>(0.0135) | 0.1108<br>(0.0137) |
| N                                                                         | 16,939,585         | 68,457             | 1,364,970          | 280,338            | 1,910,440          | 2,262,149          | 2,583,088          | 530,737            | 530,783            | 561,934            |
| <b>Panel C: Urban, Controls, and Market Access</b>                        |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |
| Urban                                                                     | 0.146<br>(0.014)   | 0.117<br>(0.014)   | 0.109<br>(0.012)   | 0.093<br>(0.011)   | 0.071<br>(0.014)   | 0.065<br>(0.016)   | 0.085<br>(0.014)   | 0.083<br>(0.015)   | 0.085<br>(0.015)   | 0.076<br>(0.018)   |
| $\log(MA)$                                                                | 0.093<br>(0.009)   | 0.079<br>(0.016)   | 0.104<br>(0.020)   | 0.089<br>(0.012)   | 0.078<br>(0.021)   | 0.186<br>(0.026)   | 0.152<br>(0.024)   | 0.147<br>(0.028)   | 0.089<br>(0.024)   | 0.074<br>(0.025)   |
| N                                                                         | 16,939,585         | 68,457             | 1,364,970          | 280,338            | 1,910,440          | 2,262,149          | 2,583,088          | 530,737            | 530,783            | 561,934            |
| <b>Panel D: Urban, Controls, and Market Access (excluding own market)</b> |                    |                    |                    |                    |                    |                    |                    |                    |                    |                    |
| Urban                                                                     | 0.254<br>(0.012)   | 0.190<br>(0.014)   | 0.194<br>(0.013)   | 0.186<br>(0.012)   | 0.134<br>(0.011)   | 0.151<br>(0.014)   | 0.147<br>(0.011)   | 0.145<br>(0.011)   | 0.122<br>(0.012)   | 0.103<br>(0.012)   |
| Remove-own $\log(MA)$                                                     | -0.020<br>(0.010)  | 0.019<br>(0.007)   | -0.034<br>(0.019)  | 0.010<br>(0.017)   | 0.007<br>(0.014)   | 0.108<br>(0.037)   | 0.102<br>(0.028)   | 0.077<br>(0.032)   | 0.059<br>(0.034)   | 0.076<br>(0.035)   |
| N                                                                         | 16,939,585         | 68,457             | 1,364,970          | 280,338            | 1,910,440          | 2,262,149          | 2,583,088          | 530,737            | 530,783            | 561,934            |

*Notes.* Each cell is the coefficient estimate corresponding to the regression for the survey year indicated by the column label. The dependent variable is log weekly earning and each regression includes an indicator for living in an urban area. Panel A just includes the Urban indicator; Panel B includes controls for age in 5-year bins, education levels, and an indicator for being White; Panel C includes the above controls and our measure of market access; Panel D modifies the  $\log(MA)$  measure by removing each unit's market access in equation ?? in Appendix subsection ?? All regressions are weighted by the IPUMS person weights. Standard errors are in parenthesis and are clustered at the MSA-Nonurban State level

## **B. MSA-level Market Access**

We aggregate our county-level measure of Market Access from Equation ?? to a measure at the urban area. For each year, we start with a crosswalk provided by IPUMS that lists which counties fall (at least partially) inside the boundaries of the metro area. Our metro area boundaries are the 1950 Standard Metropolitan Area for 1940 and 1950, the Metropolitan Statistical Areas for their respective year in 1960-2010, and the Core-Based Statistical Area for 2015.

This list is modified slightly to include MSAs that are recorded in the IPUMS survey data but are not included in that year's crosswalk. The counties that the MSA covers are identified by hand using the shape files from NHGIS for that year or if missing from the previous year.

For a given year, a metro area's market access is the geometric average of the corresponding counties' market access. Then, for our measure of market access for the nonurban portion of a state is the geometric average of all counties that are not within any metro areas. We include robustness checks for weighting the average by county population.

### **B.1 Robustness Checks**

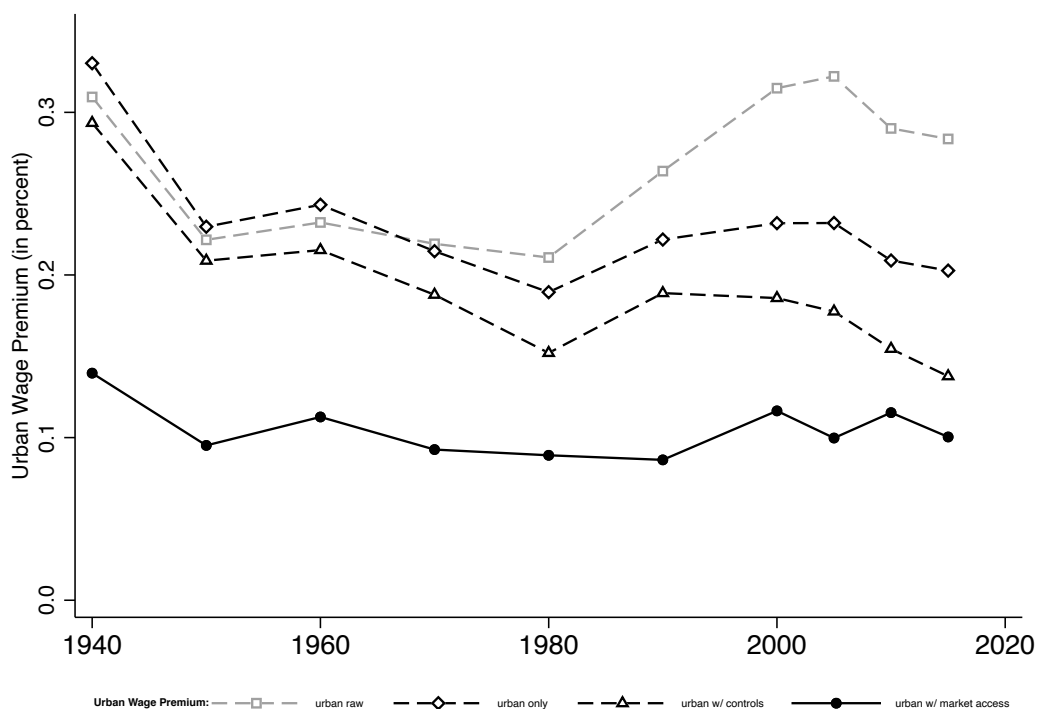
#### **B.1.1 MSA Selection Over Time**

Over decades as places grew in size, more and more places became classified as urban areas and these areas tended to be smaller than long-standing MSAs. There could be concern that the observed decline in the urban wage premium as smaller cities are added over time, the observed decline in the urban wage premium is driven by the addition of smaller cities over time. Figure B.1 shows this is not the case.

To create this, we keep all MSAs that exist in the entire sample from 1940 -

2015 (removing those that are added over time and the few that are lost over time). Observations are dropped for individuals identified in an MSA that does not exist for the entire sample. There still remains the issue that MSAs tend to grow over time incorporating less densely populated areas over time which we are unable to fix because the survey data does not feature exact locations of individuals.

Figure B.1: Urban Wage Premium 1940-2015 for 1940 MSAs



*Notes:* This figure plots the coefficient estimates for survey-year level estimates of the urban wage premium. In the grey line with square markers, the estimate is found by taking the ratio of the average log weekly wage for urban and nonurban workers. The remaining lines run versions of the regression given by Equation 2. In the line with diamond markers, there are no controls and no measure of market access; the line with triangle markers includes controls for age in 5-year bins, education levels, and an indicator for being White; the line with circle markers includes the same controls and the log of the weighted market access.